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Multi-high-gain Observer design for a class of non uniformly observable systems

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Abstract: This paper proposes a new observer design approach for a class of nonlinear systems which are not observable for all inputs. This provides an alternative to the already available solution for such systems based on Kalman with high gain, by relying here on several high gains instead of the standard single one. This can result in less conservative gains on the one hand, and excitation conditions ensuring convergence more easily checked on the other hand. A related convergence analysis is provided, and a simulation example illustrates the approach.

Keywords: Observer design, nonlinear systems, non uniformly observable systems, high gain.

1. INTRODUCTION

Observer design is known to be a topic of crucial interest, whether for control purposes or for state monitoring, and is often still a practical issue in the case of nonlinear systems. Several solutions can already be found, depending on the class of systems, as highlighted e.g. in (Bernard, 2019; Besançon, 2007, ...). Among them, high gain techniques have been shown to be pretty efficient, in particular for the case of *uniformly observable systems* (Gauthier et al., 1992). A counterpart is their high sensitivity to measurement noise (see (Khalil and Praly, 2014) for instance).
In our recent work (Besançon, 2023), a new approach has been proposed relying on several *high poles* instead of a single high gain. Noting that similar high gain techniques have also been extended to some systems which are not uniformly observable - w.r.t. inputs (Besançon and Ţiclea, 2007; Torres et al., 2012; Farza et al., 2017), the idea in the present paper is to propose a new observer design for such systems, following the same seminal idea as the one in (Besançon, 2023).
This can result in lower gains, and consequently lower sensitivity to noise, as an alternative - or complementary approach, to possible filtering solutions (Robles-Magdaleno et al., 2020). But this also results in weaker conditions on the inputs for the observer convergence, for which previously available options are pretty demanding, and have already motivated efforts in that regard (Dufour et al., 2012).
The main result here is presented in two steps: first for a special class of systems directly extending the form for usual high gain observer, and then for a more general structure.

The remainder of the paper is organized as follows: section 2 introduces the considered class of systems, while section 3 presents the proposed observer approach. A short discussion about it is given in section 4, and an illustrative example follows in section 5. Some conclusions end the paper in section 6.

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2. CLASS OF SYSTEMS

Let us consider systems which can be described by a state space representation as follows:

$$\begin{aligned} \dot{x}(t) &= A(u(t))x(t) + B(x(t),u(t)) \\ y(t) &= Cx(t) \end{aligned}$$

with

$$A(u) = \begin{pmatrix} 0 & A_{12}(u) & 0 & \cdots & 0 \\ 0 & 0 & A_{23}(u) & & \vdots \\ \vdots & & & \ddots & 0 \\ & & & & A_{q-1q}(u) \\ 0 & \cdots & & \cdots & 0 \end{pmatrix} \tag{1}$$

$$C = \begin{pmatrix} 1 & 0 & \cdots & \cdots & 0 \end{pmatrix}$$

for $u \in I\!R^m$, x made of subvectors $x_i \in I\!R^{n_i}$, $i = 1,..q$, ($n_1 = 1$) and B made of related subvectors B_i only depending on x_j's for $j \leq i$ as follows:

$$B_i(x,u) = B_i(x_1, \cdots, x_i, u) \tag{2}$$

Such a system is in general not uniformly observable in the sense that it may lose observability for some inputs. Let us further assume that:

($A1$) for $i = 2$ to q, $A_{i-1i}(u(t))$'s are uniformly bounded;
($A2$) for $i = 1$ to q, B_i's are Lipshitz functions w.r.t. x, uniformly in u.

It is known that under stronger assumptions on A_{ij}'s, one can design a global exponential observer for the system (in the sense that for any initial conditions both for the system and for the observer, the estimation error exponentially decays to zero). In particular, when they are all scalar and constant, a standard *high gain observer* (Gauthier et al., 1992) applies. The same holds true when they remain regularly away from zero (Deza et al., 1993), while a combined Kalman / high gain design is available provided u satisfies an appropriate excitation condition (Besançon and Ţiclea, 2007), corresponding to so-called *locally regular* inputs. This notion can be recalled e.g. as follows:
(LR) u is locally regular for system (1) if there exists $\lambda^* > 0$ such that $\forall \lambda \geq \lambda^*$, $\forall t \geq \frac{1}{\lambda}$:

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$$\int_{t-\frac{1}{\lambda}}^{t} \Phi_u^T(\tau,t)C^TC\Phi_u(\tau,t)d\tau \geq \alpha_0\lambda\Lambda^{-2}(\lambda) \quad (3)$$

where $\alpha_0 > 0$ is a constant, $\Lambda(\lambda)$ is a block diagonal matrix of entries $\lambda^i I_{n_i\times n_i}$, $i = 1,..q$, and Φ_u is the transition matrix for $d\xi/d\tau = A(u(\tau))\xi(\tau)$ (here, and in all the paper, $I_{r\times r}$ stands for the $r \times r$ identity matrix).
Under (LR) condition, an observer for (1)-(2) with $(A1)$-$(A2)$ satisfied, can be given by:

$$\begin{aligned} \dot{\hat{x}}(t) &= A(u(t))\hat{x}(t) + B(\hat{x}(t),u(t)) \\ &\quad -\Lambda(\lambda)K(t)(C\hat{x}(t) - y(t)) \\ K(t) &= S^{-1}(t)C^T \\ \dot{S}(t) &= \lambda(-\gamma S(t) - A^T(u(t))S(t) - S(t)A(u(t)) + C^TC) \\ S(0) &> 0 \end{aligned} \quad (4)$$

for $\gamma \geq 2\|A(u)\|$ and λ large enough.
In the present paper, we will instead consider conditions as follows for an alternative observer solution:

- $(A3)$ for $i = 2$ to q, $A_{i-1i}(u(t))'s$ are differentiable in time, with uniformly bounded derivatives,
- $(A4)$ for $i = 2$ to q, $\phi_i(u) := A_{12}(u(t))A_{23}(u(t))..A_{i-1i}(u(t))$ satisfies some (LR) condition as:

$$\exists\lambda_i^* > 0 : \forall\lambda_i \geq \lambda_i^*,\, t \geq \frac{1}{\lambda_i},\quad \int_{t-\frac{1}{\lambda_i}}^{t} \phi_i^T(u)\phi_i(u) \geq \frac{\alpha_i}{\lambda_i} \quad (5)$$

for some constant $\alpha_i > 0$ independent of λ_i.

It is a priori clear that condition $(A4)$ should be more easily checked than (3) of (LR). This will be discussed a bit further in section 4.
Notice that for the sake of easier explanations, the whole presentation will be limited to systems (1)-(2) with $q \leq 3$.

3. MULTI HIGH GAIN OBSERVER DESIGN

3.1 Case of scalar blocks in $A(u)$

Let us first consider in this subsection the case of model (1) where $n_i = 1$ for all i's, i.e. $A_{12} = a_{12} \in I\!R$ and $A_{23} = a_{23} \in I\!R$.
Following the idea of (Besançon, 2023) for the case of constant matrix $A(u)$, we can propose here a candidate observer of the following form:

$$\begin{aligned} &\dot{\hat{x}}(t) = A(u(t))\hat{x}(t) + B(\hat{x}(t),u(t)) - K(t)(C\hat{x}(t) - y(t)) \\ &\text{with} \\ &K(t) = \\ &\begin{pmatrix} \lambda_1 + \lambda_2 a_{12}^2(u(t)) + \lambda_3 a_{12}^2(u(t))a_{23}^2(u(t)) \\ \lambda_1\lambda_2 a_{12}(u(t)) + \lambda_1\lambda_3 a_{12}a_{23}^2(u(t)) + \lambda_2\lambda_3 a_{12}^3 a_{23}^2(u(t)) \\ \lambda_1\lambda_2\lambda_3 a_{12}^3(u(t))a_{23}(u(t)) \end{pmatrix} \end{aligned} \quad (6)$$

It can be noticed that when $a_{12} = a_{23} = 1$, we exactly recover the *high poles* observer of (Besançon, 2023).
Our claim at this point is that the convergence can be obtained under assumptions $(A1)$ to $(A4)$:

Theorem 1. Assume that system (1)-(2) with $q = 3$ and $n_1 = n_2 = n_3 = 1$ satisfies properties $(A1)$, $(A2)$, $(A3)$ with $a_{12}(u(t))$, and $a_{12}(u(t))a_{23}(u(t))$ LR as in $(A4)$, then there exist $\lambda_i^* > 0$ for $i = 1$ to 3 such that for any $\lambda_i \geq \lambda_i^*$, system (6) is a global exponential observer for (1)-(2).
In addition, the larger λ_i's are, the faster the convergence is. •

Using a similar trick as in (Besançon, 2023) in order to relax more the demand on gains λ_i, we also have the following variant:

Corollary 1. The same result as theorem 1 holds when $B(\hat{x},u)$ in observer (6) is replaced by $B(\bar{x},u)$ where $\bar{x}$ is defined by:

$$\bar{x} := \begin{pmatrix} \hat{x}_1 \\ \hat{x}_2 - a_{12}(\lambda_2 + \lambda_3 a_{23}^2)(\hat{x}_1 - y) \\ \hat{x}_3 - \lambda_2\lambda_3 a_{12}^3 a_{23}(\hat{x}_1 - y) \end{pmatrix} \quad (7)$$

•

The proof of theorem 1 following the same lines as the one for corollary 1, it will only be given here for the latter, since mathematical expansions are a bit reduced in that case.
For this proof, we need the following lemma:

Lemma 1. Under assumption $(A4)$, for any $\lambda_i \geq \lambda_i^*$, there exist p_i for $i = 2, 3$ sastisfying:

$$\begin{aligned} -\dot{p}_i &= -2a_i\lambda_i p_i + \lambda_i q_i, \\ \alpha_i \leq p_i &\leq \beta_i \end{aligned} \quad (8)$$

for $a_2 := a_{12}^2$, $a_3 := (a_{12}a_{23})^2$ and constants $\beta_i > \alpha_i > 0$ independent of λ_i's.

Proof. The result comes from the fact that $(A4)$ means that $\bar{a}_i(t) := a_i(\frac{t}{\lambda_i})$ is *persistently exciting* (PE) in the following sense:

$$\forall t \geq 1, \quad \int_{t-1}^{t} \bar{a}_i(\tau)d\tau \geq \alpha_i \quad (9)$$

This is indeed a direct consequence of:

$$\int_{t-1}^{t} \bar{a}_i(\tau)d\tau = \int_{\frac{t}{\lambda_i}-\frac{1}{\lambda_i}}^{\frac{t}{\lambda_i}} \lambda_i a_i(s)ds, \quad (10)$$

which is larger than α_i for $\frac{t}{\lambda_i} \geq \frac{1}{\lambda_i}$ by $(A4)$.
From (PE) property (9), the origin of the following systems is exponentially stable (see e.g. (Jenkins et al., 2018)):

$$\dot{z}_i = -\bar{a}_i z_i, \quad i = 2, 3 \quad (11)$$

Thus, by converse Lyapunov result (Khalil, 1996), given $q_i > 0$ for $i = 2, 3$, there exist $\bar{p}_i$'s $(i = 2, 3)$ satisfying the following properties:

$$\begin{aligned} -\dot{\bar{p}}_i &= -2\bar{a}_i\bar{p}_i + q_i, \\ \alpha_i \leq \bar{p}_i &\leq \beta_i \end{aligned} \quad (12)$$

for constants $\beta_i > \alpha_i > 0$. From this, there finally indeed exist $p_i(t) = \bar{p}_i(\lambda_i t)$ satisfying (8). △
We can now justify theorem 1.

Proof of theorem 1. Let us first define the following error variables:

$$\begin{aligned} e_1 &:= \hat{x}_1 - x_1 \\ e_2 &:= \hat{x}_2 - x_2 - a_{12}(\lambda_2 + \lambda_3 a_{23}^2)e_1 \\ e_3 &:= \hat{x}_3 - x_3 - \lambda_3 a_{12}^2 a_{23}e_2 - \lambda_2\lambda_3 a_{12}^3 a_{23}e_1 \end{aligned}$$

where arguments of $a'_{ii+1}s$ are omitted for lighter writing.
Then, using notations ΔB_i for $B_i(\bar{x},u) - B_i(x,u)$, we get:

$$\dot{e}_1 = -\lambda_1 e_1 + \Delta B_1 + a_{12} e_2 \tag{13}$$

$$\begin{aligned}\dot{e}_2 &= -\lambda_2 a_{12}^2 e_2 \\ &\quad + \Delta B_2 - a_{12}(\lambda_2 + \lambda_3 a_{23}^2)\Delta B_1 + a_{23} e_3 \\ &\quad -(\lambda_2 \dot{a}_{12} + \lambda_3 \dot{\overbrace{a_{12} a_{23}^2}})e_1\end{aligned} \tag{14}$$

$$\begin{aligned}\dot{e}_3 &= -\lambda_3 (a_{12} a_{23})^2 e_3 \\ &\quad + \Delta B_3 - \lambda_3 a_{12}^2 a_{23} \Delta B_2 + \lambda_3^2 (a_{12} a_{23})^3 \Delta B_1 \\ &\quad - \lambda_3 \dot{\overbrace{a_{12}^2 a_{23}}} e_2 \\ &\quad + \lambda_3 a_{12}^2 a_{23}(\lambda_2 \dot{a}_{12} + \lambda_3 \dot{\overbrace{a_{12} a_{23}^2}})e_1 - \lambda_2 \lambda_3 \dot{\overbrace{a_{12}^3 a_{23}}} e_1\end{aligned} \tag{15}$$

Notice that if time derivatives of a_{12} and a_{23} are known, terms (14) and (15) could even be directly compensated by the observer correction term (giving rise to another variant of (6)). If not, they can be upper bounded by terms of the form $c_1(\lambda_2, \lambda_3)|e_1|$ and $c_2(\lambda_2, \lambda_3)|e_1|$ respectively.
Now using lemma 1, we can consider a candidate Lyapunov function for error system (13)-(15) of the following form:

$$V := \sum_{i=1}^{3} p_i e_i^2 \tag{16}$$

where $p_1 = 1/2$, and p_2, p_3 satisfy (8).
This function is clearly positive definite.
In addition, using notations γ_{ij} for Lipshitz constants of B_i's w.r.t. x_j (uniformly w.r.t other arguments of B_i), M_{ij} for upper bounds on a_{ij}'s, and $q_1 := 1$, we can get:

$$\begin{aligned}\dot{V} \leq & -\sum_{i=1}^{3} (\lambda_i q_i - 2\gamma_{ii} p_i)\, e_i^2 \\ & + 2 \sum_{\substack{1 \leq j,k \leq 3 \\ j \neq k}} |e_j||e_k| p_j \gamma_{jk} \\ & + 2M_{12}|e_1||e_2|p_1 + 2M_{23}|e_2||e_3|p_2 \\ & + 2M_{12}(\lambda_2 + \lambda_3 M_{23}^2)\gamma_{11}|e_1||e_2|p_2 \\ & + 2\lambda_3 M_{12}^2 M_{23}|e_3|(\gamma_{22}|e_2| + \gamma_{21}|e_1|)p_3 \\ & + 2\lambda_3^2 (M_{12} M_{23})^3 \gamma_{11}|e_3||e_1|p_3 \\ & + 2\lambda_3 M_{12}^2 M_{23}|e_3||e_2|p_3 \\ & + 2\lambda_3 \max(\dot{\overbrace{a_{12}^2 a_{23}}})\gamma_{33}|e_3||e_2|p_3 \\ & + 2c_1(\lambda_2, \lambda_3)|e_1||e_2|p_2 + 2c_2(\lambda_2, \lambda_3)|e_1||e_3|p_3\end{aligned} \tag{17}$$

where, by convention, $\gamma_{jk} = 0$ whenever B_j does not depend on x_k. It is clear that quadratic terms in e_i can become negative definite for λ_i's large enough.
On the other hand, it can be checked that the right-hand side of (17) is a quadratic form $-e^T Q e$ where $e = \begin{pmatrix} e_1 & e_2 & e_3 \end{pmatrix}^T$ and Q has a stucture as follows:

$$Q = \left(\begin{array}{c|cc} \multicolumn{1}{c|}{Q_1(\lambda_1)} & \multicolumn{2}{c}{Q_{123}(\lambda_2, \lambda_3)} \\ \hline Q_{123}^T(\lambda_2, \lambda_3) & Q_2(\lambda_2) & Q_{23}(\lambda_3) \\ & Q_{23}(\lambda_3) & Q_3(\lambda_3) \end{array}\right) \tag{18}$$

where Q_i's are increasing with λ_i's.
From this, using Schur complement and determinant by blocks, one can always find $\lambda_1, \lambda_2, \lambda_3$ such that Q becomes positive definite (typically with $\lambda_1 > \lambda_2 > \lambda_3$). In addition, Q can be made larger than any diagonal matrix, meaning that $\dot{V}$ can become lower than any $-\delta e^T e$, by choosing λ_i's large enough. By standard Lyapunov arguments, the conclusion follows. △

Notice that another observer gain can be obtained when λ_2, λ_3 are roughly replaced by varying terms $\lambda_i/s_i(t)$ where $s_i(t)$ is solution of:

$$\dot{s}_i(t) = -\lambda_i \gamma_i s_i(t) + \lambda_i a_{ii+1}^2(u(t)) \tag{19}$$

for $\gamma_i > 0$, in a similar fashion as in observer (4).
Details are omitted here since they will be given in the next subsection for the more general case of rectangular blocks in $A(u)$, where we will choose indeed this alternative route for a better comparison with (4) - even if the result of theorem (1) can also be extended to this rectangular case.

3.2 General case of rectangular blocks in $A(u)$

For the more general case, we propose an observer in the same spirit as before, but now including additional time-varying elements in the gains, under the following form:

$$\begin{aligned}&\dot{\hat{x}}(t) = A(u(t))\hat{x}(t) + B(\hat{x}(t), u(t)) - K(t)(C\hat{x}(t) - y(t)) \\ &\text{with} \\ &K(t) = \begin{pmatrix} K_1(t) & K_2(t) & K_3(t) \end{pmatrix}^T \\ &K_1(t) = \lambda_1 + \lambda_2 A_{12} S_2^{-1} A_{12}^T + \lambda_3 A_{12} A_{23} S_3^{-1} A_{23}^T A_{12}^T \\ &K_2(t) = \lambda_1 \lambda_2 S_2^{-1} A_{12}^T + \lambda_1 \lambda_3 A_{23} S_3^{-1} A_{23}^T A_{12} \\ &\qquad + \lambda_2 \lambda_3 A_{23} S_3^{-1} A_{23}^T A_{12}^T A_{12} S_2^{-1} A_{12}^T \\ &K_3(t) = \lambda_1 \lambda_2 \lambda_3 S_3^{-1} A_{23}^T A_{12}^T A_{12} S_2^{-1} A_{12}^T \\ &\dot{S}_2 = -\lambda_2 \gamma_2 S_2 + \lambda_2 A_{12}^T A_{12}, \quad S_2(0) > 0 \\ &\dot{S}_3 = -\lambda_3 \gamma_3 S_3 + \lambda_3 A_{23}^T A_{12}^T A_{12} A_{23}, \quad S_3(0) > 0\end{aligned} \tag{20}$$

Notice that if A_{12} and A_{23} are scalar, we recover observer (6) by setting $S_2 := 1$, $S_3 := 1$.
Notice also that due to similarity with observer (4), it can be easily checked, for instance as in (Besançon and Ţiclea, 2007), that under condition $(A4)$ there exist $0 < \alpha_i \leq \beta_i$ independent of λ_i's such that for $i = 2, 3$ and any $t \geq \frac{1}{\lambda_i}$:

$$\alpha_i I_{n_i \times n_i} \leq S_i(t) \leq \beta_i I_{n_i \times n_i} \tag{21}$$

With this, we have the following extension of theorem 1:
Theorem 2. Under assumptions (A1), (A2), (A3), (A4), there exist $\lambda_i^* > 0$ for $i = 1$ to 3 such that for any $\lambda_i \geq \lambda_i^*$, system (20) is a global exponential observer for (1)-(2).
In addition, the larger λ_i's are, the faster the convergence is. •

As in corollary 1, we can also replace $B(\hat{x}, u)$ by $B(\bar{x}, u)$ with $\bar{x}$ given by:

$$\bar{x} := \begin{pmatrix} \hat{x}_1 \\ \hat{x}_2 - (\lambda_2 S_2^{-1} A_{12}^T + \lambda_3 A_{23} S_3^{-1} A_{23}^T A_{12})(\hat{x}_1 - y) \\ \hat{x}_3 - \lambda_2 \lambda_3 S_3^{-1} A_{23}^T A_{12}^T A_{12} S_2^{-1} A_{12}^T (\hat{x}_1 - y) \end{pmatrix} \tag{22}$$

Proof. The proof goes as before, starting now with the following definitions:

$$\begin{aligned}e_1 &:= \hat{x}_1 - x_1 \\ e_2 &:= \hat{x}_2 - x_2 - (\lambda_2 S_2^{-1} A_{12}^T + \lambda_3 A_{23} S_3^{-1} A_{23}^T A_{12}) e_1 \\ e_3 &:= \hat{x}_3 - x_3 - \lambda_3 S_3^{-1} A_{23}^T A_{12}^T A_{12} e_2 \\ &\quad -\lambda_2 \lambda_3 S_3^{-1} A_{23}^T A_{12}^T A_{12} S_2^{-1} A_{12}^T e_1\end{aligned} \tag{23}$$

We then consider the error system in e_i's and a Lyapunov function as:

$$V := e_1^2 + e_2^T S_2 e_2 + e_3^T S_3 e_3 \tag{24}$$

By computing $\dot{V}$ and using upper bounds, we can end up with $\dot{V}$ negative definite in a very similar way as in the proof of theorem 1. See appendix A for more computational details.

4. DISCUSSION

Let us here comment a bit more on observers (6) and (20). Notice first that, as in Besançon (2023), gains λ_i can be expected to result in lower values than in the case of a single one used in a high gain fashion. In particular, the form of error equations may allow to use nonlinearities in favour of convergence instead of dominating them.
In addition, the on-line computational burdern for observers (6) or (20) is obviously lower than in solution (4) for instance.
It is also important to notice that requirement $(A4)$ is in general weaker than (LR) version (3):

Proposition 1. If u is locally regular for system (1) in the sense of (LR), then condition $(A4)$ holds true.

Proof. Let us still consider $q = 3$. It can be checked that:

$$\begin{aligned} C\Phi_u(\tau,t) &= \Bigg(1 \quad \int_t^{\tau} A_{12}(u(s))ds \\ &\qquad \int_t^{\tau} A_{12}(u(s)) \int_t^{s} A_{23}(u(r))drds\Bigg) \\ &=: (\, 1 \quad \varphi_2 \quad \varphi_3 \,) \end{aligned} \tag{25}$$

Condition (LR) then clearly implies:

$$\int_{t-\frac{1}{\lambda}}^{t} \varphi_2^T\varphi_2 \geq \frac{\alpha_0}{\lambda^3} I_{n_2\times n_2} \tag{26}$$

$$\int_{t-\frac{1}{\lambda}}^{t} \varphi_3^T\varphi_3 \geq \frac{\alpha_0}{\lambda^5} I_{n_3\times n_3} \tag{27}$$

From (26), for any $z \in I\!R^{n_2}$, we have $\int_{t-\frac{1}{\lambda}}^{t}(\varphi_2 z)^2 \geq \frac{\alpha_0}{\lambda^3}z^2$. Using Cauchy Schwartz inequality on $(\varphi_2 z)^2$, and the fact that $\int_\tau^t (A_{12}z)^2 ds \leq \int_{t-\frac{1}{\lambda}}^{t}(A_{12}z)^2 ds$ for $\tau \in [t-\frac{1}{\lambda}, t]$ we finally obtain:

$$\frac{1}{2\lambda^2}\int_{t-\frac{1}{\lambda}}^{t} (A_{12}(u(\tau))z)^2 \geq \frac{\alpha_0}{\lambda^3}z^2$$

which in turn means that A_{12} satisfies $(A4)$.
Similarly, by iterated use of Cauchy Schwartz inequality, we can obtain from (27) that $A_{12}A_{23}$ also does so. △

It can further be expected that the weaker condition of *regular inputs* as introduced in Dufour et al. (2012) can also be considered in the framework of this paper, giving rise to even less restrictive conditions.

In addition, it is easy to see from the proofs exhibited in section 3, that the approach straightly extends to systems (1)-(2) where $A_{ii+1} = A_{ii+1}(u, y)$, whenever persistent excitation is satisfied as in $(A4)$, that an extension to Multi-Output case is obvious as well (just considering $n_1 > 1$), that the case $C = (\, C_1(u) \quad 0 \quad \cdots \quad 0\,)$ can be handled in a similar fashion, and that output injection can also be used.

It can finally be noticed that the design presented in this paper for the class of systems (1) with additive nonlinearity B also applies to the class of so-called state affine systems, where B reduces to a linear function. This provides an alternative to standard Kalman-like observers for such systems. This may also allow to relax condition $(A4)$ for convergence of an observer (20) in the case when some parts of B are linear: in such a case indeed, the standard Persistent Excitation can be enough instead of (LR) for the related gain design. This will be seen in the example.

5. ILLUSTRATIVE EXAMPLE

As a direct illustration of the proposed observer design, let us consider here the case of a Van der Pol oscillator, for which the approach of (Besançon, 2023) already proved to be of better performance than the standard high gain design, when writing it as (1) with unitary scalar A_{ii+1}'s. Let us assume here that the system is further subject to a control input u via some unknown gain b, and to some constant disturbance d, not known either, giving rise to the following differential description:

$$\ddot{z}(t) + \alpha^2 z(t) + \beta(1 - z^2(t))\dot{z}(t) + bu(t) + d \tag{28}$$

Considering the only measurement of z, the system can be re-written as:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\alpha^2 x_1 + \beta(1 - x_1^2)x_2 + bu + d \\ \dot{b} &= 0 \\ \dot{d} &= 0 \\ y &= x_1 \end{aligned} \tag{29}$$

which is clearly of the form (1) with $q = 3$, $A_{12} = 1$ and $A_{23} = (\, u \quad 1\,)$, while $B_1 = 0$, $B_2 = -\alpha^2 x_1 + \beta(1 - x_1^2)x_2$ and $B_3 = (\, 0 \quad 0\,)^T$ (we do not use any output injection in this example, even if this would be possible, for the sake of illustration of theorem 2).
In this configuration, the standard high gain observer does not apply any more. Solution (4) could be tried, provided that an input satisfying (LR) can be found.
The result of previous section instead allows to take advantage of the fact that in this example, only A_{23} is input-dependent, reducing the excitation requirement to $A_{23}^T A_{23}$. In particular, since B_3 is zero here, and in view of the structure of error equations (A.1), a simple PE condition is enough, meaning that a sine wave for u can do the job. On the other hand, since A_{12} is constant, we can take the steady state solution for S_2, that is e.g. $S_2 = 1$. This simplifies a lot the form of observer gain in (20).
For numerical simulation, we consider the following values:

$$\begin{aligned} &\alpha = 1,\ \beta = 1/4,\ b = 1, \\ &d(t) = 0 \text{ if } t \leq 20, \text{ and } -1/2 \text{ otherwise.} \end{aligned} \tag{30}$$

A typical related state trajectory is shown in figure 1.

For the oberver design, we will consider that state variables remain bounded as in this figure 1:

$$\forall t \geq 0, \quad |x_i(t)| \leq 4,\ i = 1, 2. \tag{31}$$

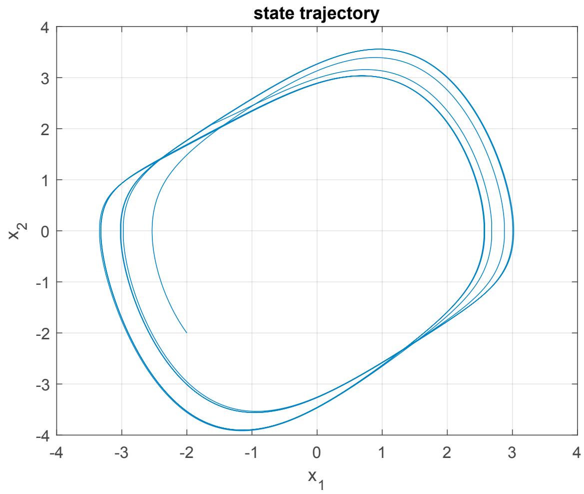


Fig. 1. A state trajectory for system (29) under sine input and step disturbance.

Gains λ_i's can then be selected by inspecting conditions for convergence of appendix A in this case, and taking advantage of elements of the dynamics in favour of observer convergence (as emphasized in (Besançon, 2023)). The following values are finally taken here:

$$\lambda_1 = 5.5,\ \lambda_2 = 4,\ \lambda_3 = 0.5 \tag{32}$$

with e.g. $\gamma_3 = 1$. The resulting state estimation errors (with observer initialization at 0) are shown in figure 2, while the input gain and disturbance estimations are displayed in figures 3 and 4 respectively.

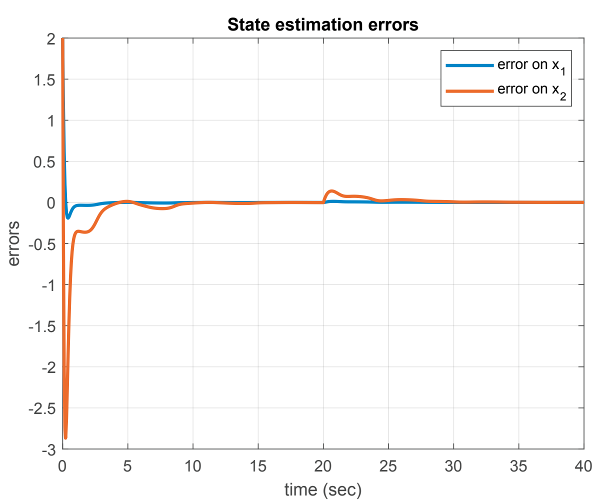


Fig. 2. State estimation errors with observer (20).

Notice that the convergence can be increased by increasing the λ_i's: estimation results for input gain b and disturbance d with all λ_i's multiplied by 3 are shown in figure 5. This is obviously at the price of increased transients.

On the other hand, keeping gains low enough can result in

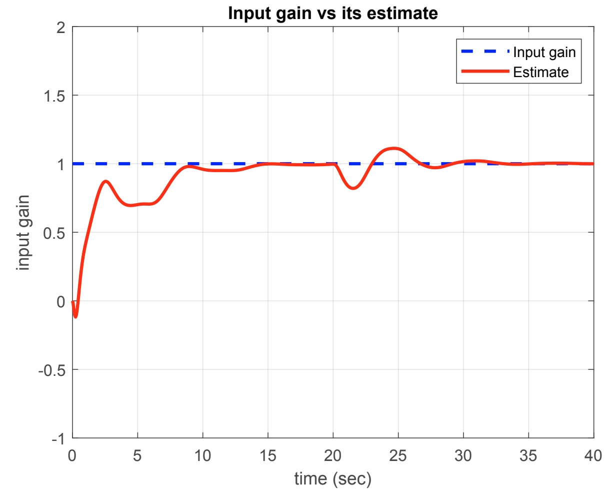


Fig. 3. Input gain estimation with observer (20).

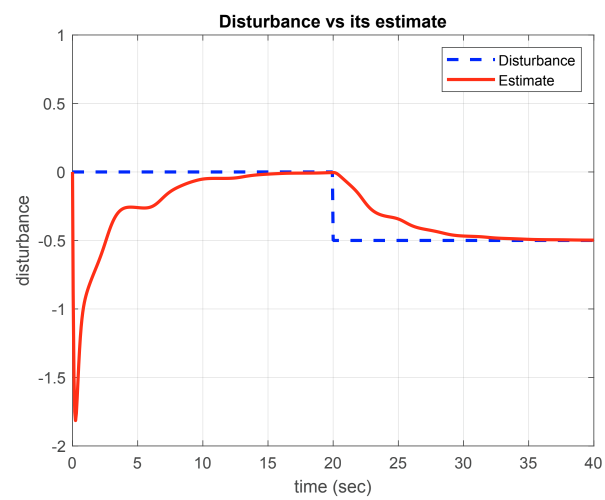


Fig. 4. Disturbance estimation with observer (20).

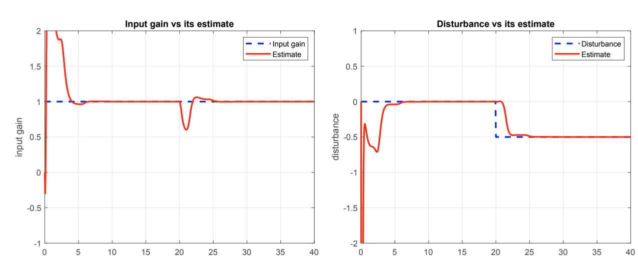
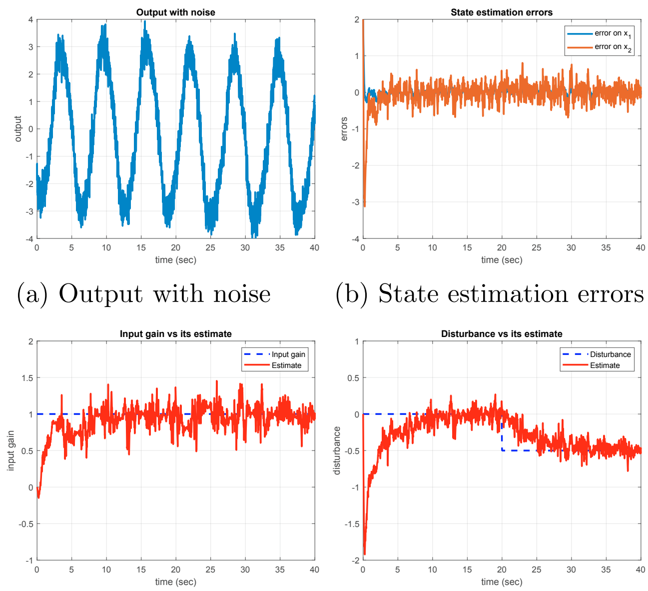


Fig. 5. Simulation results with λ_i's $\times 3$ (input gain estimation on the left, disturbance estimation on the right).

fairly good estimation performance even in the presence of measurement noise, as illustrated by figure 6 herefater (where the original tuning has been applied).

For comparison with a single high gain approach, let us recall that for the similar problem of (Besançon, 2023) where a standard high gain observer could be applied (without estimation of b, d), the theoretical gain was to be



(a) Output with noise (b) State estimation errors

(c) Input gain estimation (d) Disturbance estimation

Fig. 6. Simulation results with output noise.

chosen larger than 30, meaning a large noise amplification. In addition, in the present configuration (with estimation of b, d), it can be checked that the considered sine signal is not enough to satisfy (LR), meaning that solution (4) cannot even be expected to work with such an input.

6. CONCLUSION

In this paper, a new observer design approach has been proposed for a class of non uniformly observable systems, allowing to relax the excitation demand, as well as correction magnitude, by splitting the gain design into several 'subgains'. A simulation example has been included to illustrate the approach, and the full observer convergence proof has been provided up to some order. We expect that it can be extended in a similar fashion to any order, and leave it for a future work.

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Appendix A. PROOF OF THEOREM 2

Considering the case of observer (20) with (22) and errors (23), we get:

$$\begin{aligned}
\dot{e}_1 &= -\lambda_1 e_1 + \Delta B_1 + A_{12} e_2 \\
\dot{e}_2 &= -\lambda_2 S_2^{-1} A_{12}^T A_{12} e_2 + \Delta B_2 \\
&\quad - (\lambda_2 S_2^{-1} A_{12}^T + \lambda_3 A_{23}^T S_3^{-1} A_{23}^T A_{12}^T \Delta B_1 + A_{23} e_3 \\
&\quad -(\lambda_2 \overbrace{S_2^{-1} A_{12}^T} + \lambda_3 \overbrace{A_{23} S_3^{-1} A_{23}^T A_{12}})e_1 \\
\dot{e}_3 &= -\lambda_3 S_3^{-1} A_{23}^T A_{12}^T A_{12} A_{23} e_3 \\
&\quad + \Delta B_3 - \lambda_3 S_3^{-1} A_{23}^T A_{12}^T A_{12} \Delta B_2 \\
&\quad + \lambda_3^2 S_3^{-1} A_{23}^T A_{12}^T A_{12} A_{23} S_3^{-1} A_{23}^T A_{12}^T \Delta B_1 \\
&\quad - \lambda_3 \overbrace{S_3^{-1} A_{23}^T A_{12}^T A_{12}} e_2 \\
&\quad + \lambda_3 S_3^{-1} A_{23}^T A_{12}^T A_{12} (\lambda_2 \overbrace{S_2^{-1} A_{12}^T}^{\cdot} + \lambda_3 \overbrace{A_{23}^T S_3^{-1} A_{23}^T A_{12}^T}^{\cdot}) e_1 \\
&\quad - \lambda_2 \lambda_3 \overbrace{S_3^{-1} A_{23}^T A_{12}^T A_{12} S_2^{-1} A_{12}} e_1
\end{aligned} \tag{A.1}$$

Taking Lyapunov function (24) (which is positive definite by property (21)), we then obtain:

$$\begin{aligned}
\dot{V} \leq\ & -2(\lambda_1 - \gamma_{11}) e_1^2 - (\lambda_2 \gamma_2 e_2^T S_2 e_2 - 2||S_2|| e_2^T e_2) \\
& -(\lambda_3 \gamma_3 e_3^T S_3 e_3 - 2||S_3|| e_3^T e_3 \\
& + \mathit{cross} - \mathit{terms}
\end{aligned} \tag{A.2}$$

where quadratic terms can be made negative definite by λ_i's large enough, and cross-terms can be upper bounded using upper bounds on S_i, A_{12}, A_{23} and their time derivatives, in such a way that $\dot{V}$ admits a quadratic upper bound $-e^T Q e$ with e and Q as (18) in the proof of theorem 1. From this again, one can always find $\lambda_1, \lambda_2, \lambda_3$ such that Q becomes positive definite, and even larger than any $\delta I_{n \times n}$ for $n = n_1 + n_2 + n_3$ and arbitrary $\delta > 0$, meaning that $\dot{V}$ can become lower than $-\delta e^T e$. This ends the proof.